

STAT 5200, Notes 9¹

Minchul Shin²

Fall 2024

Readings. Finish reading Chapter 5 of W. These notes address Sections 5.2.3, 5.2.4 and 5.2.5. In the next set of notes I'll present some final comments about 2SLS estimation and turn to the topics in Chapter 6 of Wooldridge, which covers some additional issues for the single equation linear model. These include regression with estimated regressors and instruments, some specification tests..

The asymptotic efficiency of 2SLS. The efficiency of an estimator may be defined in terms of its asymptotic variance. Efficiency is a comparative statement. Let's consider a scalar parameter and a class of estimators of this parameter. For example, the class might be all unbiased estimators that are linear functions of the observed data. If we have an achievable lower bound for the asymptotic variance of all estimators in the class, then we can define the efficiency of an estimator as the ratio that is the lower bound divided by the asymptotic variance of the estimator. Thus, efficiency varies between 0 and 1. A classical example of an achievable lower bound for some parameters is the Cramér-Rao lower bound, which is equal to the reciprocal of the Fisher information for the parameter.

We can also consider the relative efficiency of two different estimators of a parameter two estimators in the same class. This can be defined as the ratio of their asymptotic variances. The estimator with the smaller asymptotic variance is said to be the one which is relatively more efficient.

Efficiency can be also be defined in terms of asymptotic mean square error, or some other measure. Where relevant, we can use the finite sample variance or the finite sample mean square error to define efficiency in the above manner. Moreover, there are other measures of efficiency of estimators in the literature.

If the parameter being estimated is a vector parameter, we can use the asymptotic covariance matrix. Then an estimator with asymptotic covariance matrix $\mathbf{V}_1$ is said to be more efficient than a second estimator with asymptotic covariance matrix $\mathbf{V}_2$ if $\mathbf{V}_2 - \mathbf{V}_1$ is positive semidefinite. Of course, we can also use mean square error in this definition.

Wooldridge proves that the 2SLS estimator is asymptotically efficient within the class of all instrumental variable estimators which are linear in the instrumental vector. Let's examine the details of the proof. Since the result is an asymptotic one, we work with population moments, which are

¹This is based on Paul Shaman's lecture notes (Notes 9 from STAT 5200) at the University of Pennsylvania. Last revised: Summer, 2024. Last compiled: August 26, 2024. Lecture notes and associated materials are licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0) license.

²Email: mincshin@wharton.upenn.edu

probability limits of sample sums used to construct the 2SLS estimator. We begin by recalling the notation used in Notes 8, and also some of the details in the construction of the 2SLS estimator. See the development in Section 5.2.1 of W.

Let's assume that Assumptions 2SLS.1–3 hold. The third of these assumptions is homoscedasticity. The model we wish to estimate is (observe the notation here)

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_{K-1} x_{K-1} + \beta_K x_K + u = \mathbf{x}\beta + u. \quad (1)$$

Assumption 2SLS.1 states that we have observed values of a $1 \times L$ vector $\mathbf{z}$ for which $E[\mathbf{z}'u] = \mathbf{0}$. Also, $L \geq K$. The vector $\mathbf{z}$ includes the exogenous explanatory variables in $\mathbf{x}$ plus the instrumental variables to be used. In the first stage of 2SLS we construct $\mathbf{x}^*$, the linear projection of $\mathbf{x}$ onto $\mathbf{z}$. This projection is

$$\mathbf{x}^* = \mathbf{z} [E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'\mathbf{x}], \quad (2)$$

where $E[\mathbf{z}'\mathbf{z}]$ is nonsingular by Assumption 2SLS.2. It follows from (2) that

$$E[\mathbf{x}^{*'}\mathbf{x}] = E[\mathbf{x}^{*'}\mathbf{x}^*] = E[\mathbf{x}'\mathbf{z}] [E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'\mathbf{x}]. \quad (3)$$

The 2SLS estimator of β is given by (22) of Notes 8,

$$\begin{aligned} \hat{\beta} &= \left(\sum_{i=1}^N \hat{\mathbf{x}}_i' \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \hat{\mathbf{x}}_i' y_i \right) \\ &= \left[\left(\sum_{i=1}^N \mathbf{x}_i' \mathbf{z}_i \right) \left(\sum_{i=1}^N \mathbf{z}_i' \mathbf{z}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}_i' \mathbf{x}_i \right) \right]^{-1} \left[\left(\sum_{i=1}^N \mathbf{x}_i' \mathbf{z}_i \right) \left(\sum_{i=1}^N \mathbf{z}_i' \mathbf{z}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{z}_i' y_i \right) \right], \end{aligned} \quad (4)$$

where $\hat{\mathbf{x}}_i$ is the OLS prediction of $\mathbf{x}_i^*$ [see (14) and (15) of Notes 8]. The asymptotic variance of the 2SLS estimator in (4), that is, of $\sqrt{N}(\hat{\beta} - \beta)$, assuming homoscedasticity, is [see (24) of Notes 8]

$$Avar \left(\sqrt{N}(\hat{\beta} - \beta) \right) = \sigma^2 \left[E[\mathbf{x}'\mathbf{z}] [E[\mathbf{z}'\mathbf{z}]]^{-1} E[\mathbf{z}'\mathbf{x}] \right]^{-1} = \sigma^2 \left[E[\mathbf{x}^{*'}\mathbf{x}^*] \right]^{-1}, \quad (5)$$

from (3) and (4).

Now let $\tilde{\beta}$ be any estimator constructed using a linear combination of the instrumental vector for $\mathbf{x}$, any combination *not defined by the linear projection* of $\mathbf{x}$ onto $\mathbf{z}$. Denote this linear combination by $\tilde{\mathbf{x}}$, so that

$$\tilde{\mathbf{x}} = \mathbf{z}\Gamma, \quad (6)$$

where Γ is an $L \times K$ matrix. Then [the analog of (5.17) in W]

$$\tilde{\beta} = \left(\sum_{i=1}^N \tilde{\mathbf{x}}_i' \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \tilde{\mathbf{x}}_i' y_i \right), \quad (7)$$

and the asymptotic variance of $\sqrt{N}(\tilde{\beta} - \beta)$ is thus

$$Avar\left(\sqrt{N}(\tilde{\beta} - \beta)\right) = \sigma^2 [E[\tilde{\mathbf{x}}'\mathbf{x}]]^{-1} E[\tilde{\mathbf{x}}'\tilde{\mathbf{x}}] [E[\mathbf{x}'\tilde{\mathbf{x}}]]^{-1}. \quad (8)$$

Recall that $\mathbf{x}^*$ is the linear projection of $\mathbf{x}$ onto $\mathbf{z}$. Thus we can write $\mathbf{x} = \mathbf{x}^* + \mathbf{r}$, where $\mathbf{r}$ is uncorrelated with $\mathbf{z}$, $E[\mathbf{z}'\mathbf{r}] = \mathbf{0}$. Then it follows from (6) that $E[\tilde{\mathbf{x}}'\mathbf{r}] = 0$. Then

$$E[\tilde{\mathbf{x}}'\mathbf{x}] = E[\tilde{\mathbf{x}}'(\mathbf{x}^* + \mathbf{r})] = E[\tilde{\mathbf{x}}'\mathbf{x}^*],$$

and thus

$$Avar\left(\sqrt{N}(\tilde{\beta} - \beta)\right) = \sigma^2 [E[\tilde{\mathbf{x}}'\mathbf{x}^*]]^{-1} E[\tilde{\mathbf{x}}'\tilde{\mathbf{x}}] [E[\mathbf{x}^*\tilde{\mathbf{x}}]]^{-1}. \quad (9)$$

It remains to show that $\hat{\beta}$ is more efficient than $\tilde{\beta}$. It suffices to show that the inverse of (5) minus the inverse of (9) is positive semidefinite, or that

$$E[\mathbf{x}^{*'}\mathbf{x}^*] - E[\mathbf{x}^{*'}\tilde{\mathbf{x}}] [E[\tilde{\mathbf{x}}'\tilde{\mathbf{x}}]]^{-1} E[\tilde{\mathbf{x}}'\mathbf{x}^*] \quad (10)$$

is positive semidefinite. But by direct multiplication (10) is equal to

$$E\left[\left(\mathbf{x}^* - \tilde{\mathbf{x}} [E[\tilde{\mathbf{x}}'\tilde{\mathbf{x}}]]^{-1} E[\tilde{\mathbf{x}}'\mathbf{x}^*]\right)' \left(\mathbf{x}^* - \tilde{\mathbf{x}} [E[\tilde{\mathbf{x}}'\tilde{\mathbf{x}}]]^{-1} E[\tilde{\mathbf{x}}'\mathbf{x}^*]\right)\right] = E[\mathbf{s}^{*'}\mathbf{s}^*], \quad (11)$$

which is positive semidefinite. Note that $\mathbf{s}^*$ is the residual from the linear projection of $\mathbf{x}^*$ onto $\tilde{\mathbf{x}}$. Remember that this efficiency result depends on the assumption of homoscedasticity.

Further comments on hypothesis testing with 2SLS. In Section 5.2.4 Wooldridge gives a residual-based method for testing a multiple linear hypothesis. In OLS analysis one can construct the F -test of a multiple linear hypothesis $H_0 : \mathbf{R}\beta = \mathbf{r}$ by using the F statistic analog of the Wald statistic, or by using a residual-based construction (equivalently, one based on an R square construction).

If one has formed the 2SLS estimator, the Wald approach can be employed using the asymptotic covariance matrix describe in Notes 8 when assuming homoscedasticity or assuming heteroscedasticity.

For the 2SLS estimator Wooldridge describes a residual-based F -statistic to test a hypothesis which is an exclusion restriction. Write the model to be estimated as

$$y = \mathbf{x}\beta + u = \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + u,$$

where $\mathbf{x}_1$ is $1 \times K_1$, $\mathbf{x}_2$ is $1 \times K_2$, $K_1 + K_2 = K$, and the hypothesis to be tested is

$$H_0 : \beta_2 = \mathbf{0} \quad vs. \quad H_1 : \beta_2 \neq \mathbf{0}.$$

Each part of the $\mathbf{x}$ vector can contain both exogenous and endogenous variables. Wooldridge's statistic is described in Section 5.2.4.

Note that the restricted model is

$$y = \mathbf{x}_1\beta_1 + u.$$

The instrumental vector is denoted by $\mathbf{z}$; it contains the exogenous explanatory variables and the instrumental variables. The test Wooldridge describes uses an F -type statistic and requires calculation of three different sums of squares. One is

$$SSR_{ur} = \sum_{i=1}^N \hat{u}_i^2, \quad \hat{u}_i = y_i - \mathbf{x}_i\hat{\beta}, \quad i = 1, \dots, N.$$

That is, $\hat{u}_i$ is the i th residual from 2SLS using the unrestricted model. Another sum of squares is

$$\widehat{SSR}_{ur} = \sum_{i=1}^N \tilde{u}_i^2, \quad \tilde{u}_i = y_i - \hat{\mathbf{x}}_i\hat{\beta}, \quad i = 1, \dots, N,$$

where $\hat{\beta}$ is the estimate of β from the second-stage regression. That is, $\tilde{u}_i$ is the i th residual from the second-stage regression using the unrestricted model. The third sum of squares uses the residuals from the second-stage regression for the *restricted* model. For this the first-stage regressions fit each endogenous explanatory variable in $\mathbf{x}_1$ vs. the vector $\mathbf{z}$ containing the exogenous explanatory variables in $\mathbf{x}_1$, $\mathbf{x}_2$ and the instrumental variables. This produces predicted values $\hat{\mathbf{x}}_1$, and the second-stage regression is y vs. $\hat{\mathbf{x}}_1$. Thus, the third sum of squares is

$$\widehat{SSR}_r = \sum_{i=1}^N \bar{u}_i^2, \quad \bar{u}_i = y_i - \hat{\mathbf{x}}_{i1}\bar{\beta}_1, \quad i = 1, \dots, N.$$

where $\bar{\beta}_1$ is the estimate of β_1 from the second-stage regression. The F -type statistic is then

$$F = \frac{(\widehat{SSR}_r - \widehat{SSR}_{ur}) / K_2}{SSR_{ur} / (N - K)}, \quad (12)$$

and this is approximately F with K_2 and $N - K$ degrees of freedom. Note that, on line 4 below (5.30) on page 105 of W, SSR_{ur} should read $\widehat{SSR}_{ur}$, and on line 5 below (5.30), SSR_r should read $\widehat{SSR}_r$.

For Wooldridge's Example 5.4 the unrestricted model is

$$\log wage \quad vs. \quad exper, \quad exper2, \quad educ, \quad kidslt6, \quad \text{and} \quad kidsge6,$$

and the restricted model is

$$\log wage \quad vs. \quad exper, \quad exper2, \quad \text{and} \quad educ.$$

We will duplicate Wooldridge's result. See the companion notes for the lab session.

Example. Wooldridge presents estimation of a wage equation for 428 married, working women using the Mroz data set. See Examples 4.1, 5.3, and 5.4. See the companion notes for the lab session.